



From crash reports to safer roads: a multimodal framework integrating vision-language models and street view analysis

Guanhe Wu^a, Xintong Yan^a, Yajie Zou^b, Yuanchang Xie^{a,*}

^a Department of Civil and Environmental Engineering, University of Massachusetts Lowell, Lowell, MA 01854, USA

^b Department of Transportation Engineering, Tongji University, Shanghai 200092, China

ARTICLE INFO

Keywords:

Road safety diagnostics
Street-view imagery
Crash narratives
Vision-language model
Root cause analysis
Large language model

ABSTRACT

Crash narratives and diagrams contain rich causal and contextual information that is often underutilized in traditional road safety analysis, which typically relies on aggregated crash counts and tabulated variables. Such approaches have limited ability to explain the mechanisms of individual crashes. This study proposes a mechanism-oriented, multimodal framework for scalable and interpretable roadway risk diagnosis. The framework employs a vision-language model to jointly analyze each crash narrative and diagram, constructing an explicit reasoning chain that distinguishes immediate crash-triggering behaviors from underlying contextual constraints. Based on inferred root cause keywords, each crash is decomposed into proportional contributions of five predefined factors: human, vehicle, road, environment, and traffic signal. These attributions are spatially aggregated to identify hotspots with elevated road-related risk. For high-risk locations, street-view imagery is analyzed in the crash context to diagnose potential roadway design deficiencies and generate targeted recommendations. The framework is evaluated using 4,302 crash reports from Massachusetts with expert-annotated attributions. Vision-language models substantially outperform locally trained regression baselines, particularly under imbalanced factor distributions. Among all tested models, the Grok 2 multimodal model achieves the highest agreement with expert annotations. Further analysis shows that the distribution of inferred root cause keywords in the factor attribution space effectively explains how relative factor contributions are formed. The street-view-based diagnostic outputs align closely with real-world engineering issues, demonstrating that combining causal reasoning with multimodal crash data enables practical, mechanism-driven road safety diagnostics beyond frequency-based screening.

1. Introduction

A thorough understanding of individual crashes is essential for developing effective interventions. In aviation, each crash is typically subjected to an in-depth investigation that examines human behavior, equipment conditions, and environmental factors, resulting in a comprehensive understanding of the causal mechanisms (Liu et al., 2025a; Aguiar et al., 2017). In contrast, road traffic crashes, due to their much higher frequency, rarely receive the same level of scrutiny. Road safety studies often rely on aggregated and structured data, including crash frequency and roadway characteristics such as segment length, lane width, and shoulder width. While this information supports large-scale statistical modeling, it fails to capture the complex interactions that contribute to individual crash events (Yan et al., 2021b).

Crash reports often contain rich unstructured information, such as

crash narratives and diagrams, which document the behavioral logic, spatial relationships, roadway conditions, and surrounding environment at the time of the crash as illustrated in Fig. 1. These elements offer valuable insights into the underlying mechanisms of individual crashes (Fitzpatrick et al., 2017). In recent years, crash narratives have received growing attention in safety analysis. Most studies have employed traditional natural language processing techniques to extract key entities from text, aiming to supplement structured and tabulated crash data. Although these efforts have improved data quality to some extent, they primarily focus on surface-level information extraction or crash-type categorization and remain largely auxiliary to statistical summaries, without explicitly modeling interactions among higher-level contributing factors or the causal structure embedded in crash narratives (Kwayu et al., 2021; Li et al., 2024; Arteaga-Sanchez, 2022).

With the development of large language models (LLMs), researchers

* Corresponding author.

E-mail addresses: guanhe_wu@student.uml.edu (G. Wu), xintong_yan@uml.edu (X. Yan), yajiezou@hotmail.com (Y. Zou), yuanchang.xie@uml.edu (Y. Xie).

<https://doi.org/10.1016/j.aap.2026.108419>

Received 24 August 2025; Received in revised form 9 January 2026; Accepted 18 January 2026

Available online 23 January 2026

0001-4575/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

have begun to explore more complex semantic interpretations of crash narratives by leveraging these models' language understanding and reasoning capabilities. LLMs can identify behavioral anomalies, conflict scenarios, and surface-level causes of crashes, marking a shift from simple information extraction toward causal reasoning (Mumtarin et al., 2023; Arteaga et al., 2024). However, existing applications are often limited to identifying individual crash contributing factors and tend to emphasize single, surface-level causes. As a result, interactions among multiple contributing factors are insufficiently captured, and the influence of roadway geometric and contextual constraints remains under-explored. For instance, crashes described as involving a "failure to yield" or "failure to decelerate" are often attributed to immediate driver actions, such as inadequate attention to surrounding traffic. However, when narrative descriptions are examined together with crash diagrams, it becomes evident that such behaviors may occur within constrained merging or channelized areas where roadway geometry, lane configuration, and signage design systematically limit drivers' perception and response options (Fig. 1). Unlike prior NLP-based or vision-only approaches that treat these behaviors as isolated driver errors or rely on generic visual feature detection, such cases require reasoning over the interaction between observed driver actions and latent roadway constraints. This tendency toward single-cause attribution reflects a lack of mechanism-level representation obscures deeper systemic risks and undermines the effectiveness of subsequent interventions, limiting the practical value of crash analysis for roadway safety improvement (Roque et al., 2019; Kwayu et al., 2021). At the same time, crash diagrams, which capture critical spatial and interactive features of crashes at the local level, have been largely overlooked in previous studies, leading to a lack of spatial structure in crash mechanism modeling.

Crash causation often results from the interplay of multiple factors, including human behavior, vehicle characteristics, roadway conditions, environmental context, and traffic control. An effective modeling of these factors, particularly in identifying spatial concentrations of

roadway-related risk, would significantly enhance the precision of safety interventions (Yan et al., 2021a). In traditional practice, engineers typically rely on on-site inspections to identify and diagnose roadway deficiencies. However, this process is time-consuming, labor-intensive, and difficult to scale. In contrast, street-view imagery has emerged as a valuable source of semantic information that reflects real-world roadway features such as geometry, lane markings, signage visibility, and roadside conditions (Ren et al., 2024). With recent advancements in vision-language models (VLMs), it is now possible to automatically analyze street-view imagery within the context of crash reports, offering a feasible solution for large-scale, remote roadway diagnostics (Yue, 2025).

Despite these advances, existing computer vision-based approaches typically perform generic feature detection on images without direct linkage to crash-specific causal mechanisms. Image information alone cannot explain why a particular crash occurred, nor how detected roadway features interact with human behavior or traffic control to produce risk.

In response to the limitations of current approaches in causal inference, spatial risk identification, and diagnostic capability, this study proposes a multi-source framework for roadway risk assessment. Rather than introducing another crash prediction or hotspot detection model, the proposed framework focuses on mechanism-level interpretation by explicitly linking crash occurrence to contributing factors through structured reasoning. In this sense, the framework directly responds to recent calls in the literature for improving model interpretability and strengthening the connection between data-driven analysis and practical safety management applications (Ali et al., 2024). The framework centers on crash narratives and diagrams to decompose each crash into five contributing factors: human, vehicle, road, environment, and traffic control, and estimate the relative contribution of each. By jointly leveraging textual descriptions and spatial cues from diagrams, the framework produces structured, mechanism-oriented factor attributions

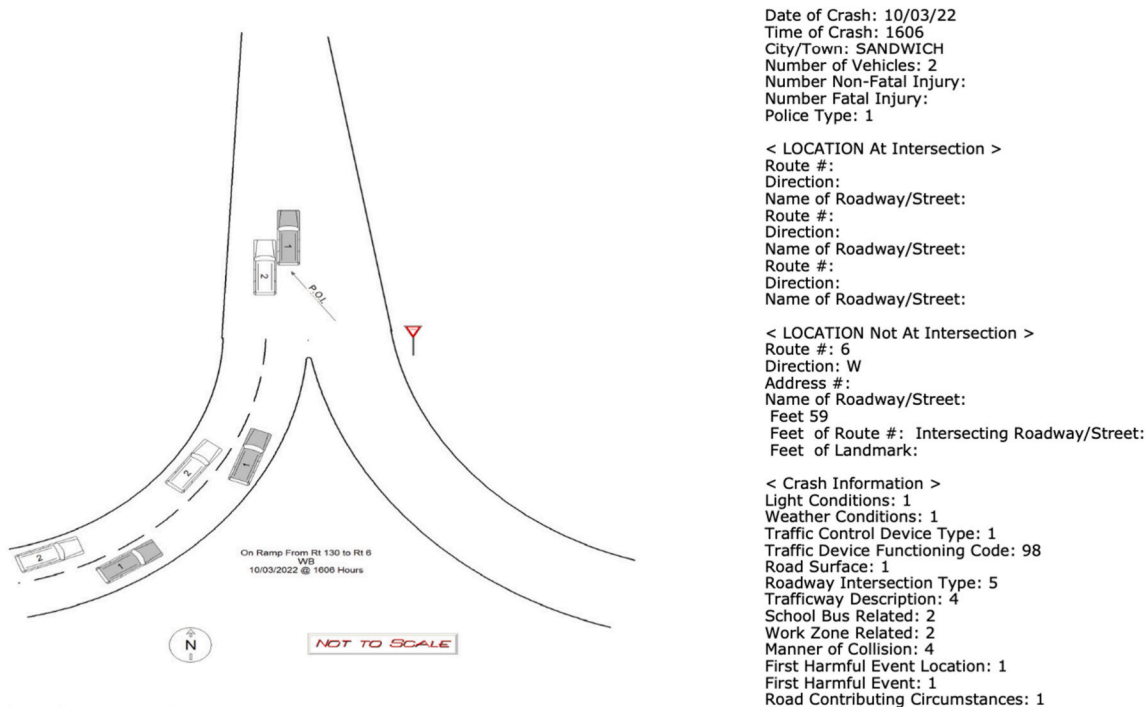


Fig. 1. Example crash report illustrating the complementary roles of narrative text and diagram in understanding crash mechanisms.

that go beyond single-modality analysis. By aggregating the spatial distribution of road-related causes, the framework identifies high-risk locations that reflect underlying systemic issues. For these locations, corresponding street-view imagery is analyzed to detect potential deficiencies in road geometry, lane markings, signage, sight distance, and roadside features. Importantly, this diagnostic process is guided by the inferred crash mechanisms, maintaining a coherent explanatory chain from crash occurrence to infrastructure condition. The results form a structured and interpretable output that supports large-scale roadway safety evaluations and infrastructure improvement planning. The main contributions of this study are as follows:

1. A multimodal causal reasoning framework based on vision-language models is proposed to jointly analyze crash narratives and diagrams. Unlike prior NLP-only or vision-only approaches that focus on surface-level extraction or classification, the proposed framework explicitly constructs a reasoning chain to infer both immediate crash-triggering behaviors and underlying contributing conditions from crash reports. These inferred root-cause keywords are then systematically synthesized into proportional contributions of five pre-defined factors (human, vehicle, road, environment, and traffic control), enabling mechanism-level interpretation at the individual crash level.
2. Building upon the factor-level attribution results, a spatial aggregation and diagnostic framework is developed to identify roadway locations with elevated road-related risk. By aggregating road-related causal contributions across space and integrating street-view imagery within the inferred crash context, the proposed approach enables interpretable, context-aware diagnosis of roadway design deficiencies and supports actionable, mechanism-driven safety interventions.

2. Literature review

2.1. From structured data to narrative-based crash analysis

Traffic crash reports have long served as a critical data source for road safety analysis. Aggregated and tabulated data derived from these reports have supported numerous safety studies employing statistical and machine learning methods to identify crash patterns and contributing factors (Afghari et al., 2019; Yan et al., 2021b; Santos et al., 2021; Al-Mistarehi et al., 2022). These approaches have proven effective for large-scale network screening and for identifying key features associated with crash occurrence. However, because they primarily rely on aggregated variables, such methods offer limited insight into the dynamic sequence of events leading to individual crashes. As a result, they fall short in uncovering the causal mechanisms of crashes.

To overcome the limitations of structured data in modeling crash mechanisms, many studies have turned to the unstructured content found in crash reports, especially the narrative sections that describe driver behavior, sequences of vehicle conflicts, and roadway conditions. Traditional natural language processing approaches typically employ keyword extraction (Sayed et al., 2021), named entity recognition (Yan et al., 2023), and dependency parsing (Zhang et al., 2024) to identify elements such as vehicle type, travel direction, traffic violations, and the initial collision position. These elements are then used to supplement structured fields or to build taxonomies of crash scenarios (Kwayu et al., 2021; Shao et al., 2024; Mumtarin et al., 2023; Arteaga et al., 2024). While such methods improve data completeness and consistency, they focus mainly on information extraction and simple classification. Consequently, they fall short of constructing the causal chains that explain how crashes unfold and therefore cannot reveal the deeper mechanisms behind crash events.

In recent years, large language models (LLMs) have demonstrated enhanced capabilities in semantic understanding and reasoning. Some studies have begun to leverage these models for more advanced text

analysis, such as identifying underlying patterns of risky behavior, contextual features of vehicle conflicts, or thematic causes of crashes (Fan et al., 2024; Arteaga and Park, 2025; Grigorev et al., 2024; Zheng et al., 2023). This shift represents an initial transition from attribute field-level completion toward mechanism-level reasoning. However, current analyses based on LLMs still face significant limitations. For each crash, the models often identify only a single, surface-level cause, such as failure to slow down or yield, without uncovering the interactions among multiple contributing factors, such as how poor road design might trigger driving errors. Furthermore, the inferred causes are typically not validated against real-world roadway conditions, resulting in limited spatial accuracy and low operational relevance. As a result, they provide limited support for precise roadway risk diagnosis.

2.2. Multimodal understanding of crash mechanisms

In addition to textual descriptions, crash diagrams also contain a wealth of critical spatial information, including the relative positions of vehicles, directions of conflict, roadway layout, and the locations of obstacles. This information plays a vital role in reconstructing the dynamic process leading to a crash. Diagrams and narratives provide complementary perspectives and together form the most informative sources of unstructured data within crash reports (Jain et al., 2024). However, compared to narratives, the use of diagrams in current research remains very limited. Existing work largely focuses on manual interpretation or basic digitization, lacking systematic extraction of visual features and deeper causal analysis based on diagram content.

Although crash narratives and diagrams together represent the most important sources of unstructured information in crash reports, they are often processed separately in existing research. This lack of a unified modeling approach limits the ability to conduct integrated analysis and causal reasoning based on multiple sources of information. To address this challenge, a unified model capable of understanding both textual and visual semantics and performing reasoning tasks is needed. In recent years, vision-language models have demonstrated strong capabilities in jointly processing images and text while performing complex reasoning tasks. These multimodal systems have shown great potential in fields such as medical image analysis (Van et al., 2024), autonomous driving perception (Zhou et al., 2024), and robotic manipulation (Gao et al., 2024). Although their application in traffic safety remains in the very early stages, some studies have begun to explore their use in roadway scene understanding (Jain et al., 2024), driver behavior prediction (Guo et al., 2024), and pedestrian behavior and safety (Gao and Zhao, 2025). Integrating these models into traffic safety research may enable multi-source causal modeling of crash events and enhance both the depth and interpretability of mechanism identification.

2.3. Street view imagery for roadway-level risk diagnosis

Beyond the information provided in crash reports, street view imagery offers a realistic and perceptually grounded foundation for road safety analysis. As a widely accessible, cost-effective, and semantically rich data source, street view imagery captures critical roadway attributes such as geometric design, lane markings, signage visibility, sight distance limitations, and roadside safety features (Wang et al., 2025; Ye et al., 2025). Prior research has shown that integrating street view imagery with deep learning models enables the effective identification of high-risk roadway characteristics, supporting crash prediction and infrastructure maintenance (Yue, 2025; Yu et al., 2024; Cai et al., 2022). In addition to representing physical features, street view imagery also reflects perceptual and emotional cues related to the built environment, which have been shown to influence drivers' risk perception and behavior (Liu et al., 2025b). Incorporating street view imagery into vision-language models allows for spatial localization, visual verification, and context-aware risk diagnosis under realistic road conditions (Tanprasert et al., 2020). This integration enhances the interpretability,

accuracy, and operational value of crash causation analysis and provides a more robust foundation for mechanism-driven road safety interventions.

3. Methodology

The proposed approach consists of two stages, as illustrated in Fig. 2, and is designed to support mechanism-level crash interpretation and roadway risk diagnosis using multimodal crash data.

In the first stage, a vision language model is employed to jointly analyze the narrative text and diagram contained in each traffic crash report. Guided by predefined role definitions, factor definitions, and reasoning rules, the model constructs a structured reasoning process rather than performing direct label prediction. Through this process, the model identifies immediate causes reflected in observed crash behaviors and infers underlying contributing conditions embedded in the roadway, traffic control, and environmental context. The relative contribution of each crash factor is determined by jointly considering evidence from both causal layers, integrating behavioral triggers and contextual constraints to assess how multiple mechanisms collectively contribute to the crash outcome. Based on this integrated reasoning, each crash is decomposed into five predefined contributing factors, including human, vehicle, road, environment, and signal control, and a normalized attribution vector is produced to represent their relative contributions.

In the second stage, the geographic coordinates extracted from the reports are used to identify spatial clusters of crashes through DBSCAN. Each crash is then associated with the factor scores from the first stage, enabling the computation of a composite risk score related to road conditions. Locations with high crash density and elevated road-related risk are identified as high-risk areas. For these locations, Google Street View images are retrieved and analyzed together with the corresponding crash texts using language models. This process supports the identification of possible road deficiencies and provides recommendations for targeted infrastructure improvements.

3.1. Stage 1: root cause attribution using multimodal language models

3.1.1. Dataset and annotation

This study is based on a dataset of 4,302 traffic crash reports

collected over four years (2018–2019, 2021–2022) from the Massachusetts Department of Transportation (MassDOT). The dataset focuses specifically on crashes that occurred at or near signal-controlled ramp terminals, which are critical locations for evaluating safety at freeway interchanges. The crashes span 16 municipalities across Massachusetts and were reported by multiple local police departments, reducing the likelihood that the dataset reflects the reporting practices of a single agency. A small number of reports with unclear or incomplete narrative text or illegible crash diagrams were excluded to ensure the reliability of subsequent analysis.

Each crash report contains detailed information, including:

- Structure information such as time and location of the crash
- A narrative description of the crash event
- A diagram illustrating vehicle positions and movements

To validate the accuracy of model-generated root cause attributions, we conducted expert annotations on a subset of 500 crash reports sampled from a total of 4,302 reports. The sample was selected to proportionally represent different geographic regions within the dataset.

Each crash was manually labeled according to the estimated contribution of five predefined categories: human, vehicle, road, environment, and traffic signal. Traffic signal factors were treated as a distinct category given the specific context of the dataset, which focuses on crashes occurring at or near ramp terminals, where signal-related issues are both common and influential. The annotation was performed independently by two trained reviewers with experience in transportation safety analysis, following a standardized annotation protocol derived from traffic safety literature and state crash investigation guidelines. A senior researcher with prior experience in crash investigation adjudicated cases with substantial disagreement. Given that the annotations consist of proportional factor attribution vectors rather than discrete class labels, inter-annotator consistency was evaluated using complementary metrics that reflect both dominant-factor agreement and overall attribution structure. Specifically, a Top-1 match rate was used to assess agreement on the dominant contributing factor, yielding a value of 0.972, indicating near-perfect consistency in identifying the primary cause of crashes. In addition, Spearman rank correlation was computed between the full attribution vectors from the two annotators, resulting in a correlation of 0.874, which

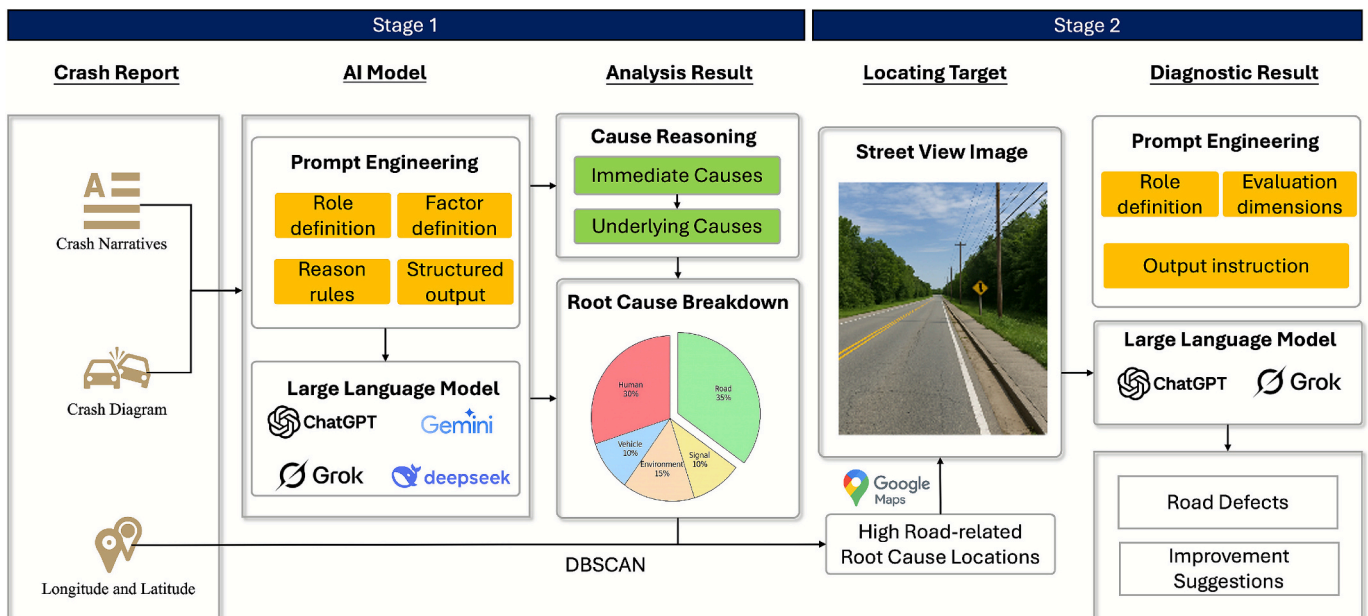


Fig. 2. Two-Stage Framework for Road Diagnostics Using Crash Reports.

demonstrates strong agreement in the relative ordering and distribution of contributing factors.

For each report, a set of scores summing to 100 percent were assigned across the five categories. For example, one annotated case yielded the attribution vector [0.40, 0.10, 0.30, 0.15, 0.05], corresponding to the five categories in order. This indicates a crash primarily related to human and road factors, with additional contributions from environmental and signal conditions.

The definitions, representative examples, and proportional scoring principles of each category are as follows:

- **Human:** factors related to driver behavior and decision-making, including distraction (such as mobile phone use), misjudgment of gaps or speeds, fatigue, failure to yield, violation of traffic rules, aggressive driving (such as speeding or tailgating), and impaired driving due to alcohol or drug use. Human factors were assigned higher proportional scores (typically ≥ 0.6) when driver behavior was identified as the primary and sufficient cause of the crash based on clear narrative or diagram evidence. When human behavior contributed jointly with other factors, moderate scores (0.2–0.4) were assigned depending on its relative role. Weakly supported or indirect behavioral influences received smaller shares (≤ 0.2).
- **Vehicle:** mechanical or functional failures of the vehicle that compromise safety, such as brake malfunction, tire blowouts, steering system defects, headlight failure, or engine problems. Vehicle factors were assigned higher scores only when explicit evidence of mechanical failure was documented or strongly implied in the report. In cases where vehicle condition played a secondary or enabling role, moderate scores were assigned, while speculative or undocumented vehicle-related issues received minimal or zero contribution.
- **Road:** issues related to the physical design, layout, or condition of the roadway, including insufficient sight distance, sharp curves, narrow lanes, faded or missing lane markings, inadequate signage, poor drainage, unsafe merging zones, and the presence of foreign objects or debris on the road surface. Road-related factors received higher proportional scores when roadway characteristics were identified as a dominant constraint shaping driver behavior or vehicle interaction. When roadway deficiencies interacted with human or environmental factors, moderate shares (0.2–0.4) were assigned to reflect joint influence, while minor or uncertain roadway issues received smaller allocations.
- **Environment:** external conditions that influence driving behavior or roadway safety, such as rain, snow, fog, ice, glare from sunlight, low visibility at night, and prevailing traffic conditions, including heavy congestion or sparse vehicle presence. Environmental factors were scored based on their observed impact on visibility, friction, or traffic dynamics. Severe or clearly influential conditions received higher scores, while mild or background environmental conditions were assigned lower proportional contributions.
- **Traffic signal:** problems involving the visibility, configuration, or timing of traffic signals, such as signals that are partially obstructed, entirely missing, improperly phased, or inconsistently placed. Traffic

signal factors were assigned higher scores when signal-related issues directly affected driver decision-making or conflict generation at ramp terminals. When signal issues were contributory but not dominant, moderate scores were assigned, and minimal scores were used when signal involvement was indirect or weakly supported.

- In crashes without a single dominant factor, multiple categories could receive moderate proportional shares depending on the combined evidence from narratives and diagrams. This integrated definition-and-scoring structure ensures that factor attribution reflects both categorical meaning and relative causal strength.

The overall distribution of factor contributions across the 500 annotated reports is shown in Fig. 3. Human factors account for the largest share among all categories, with more than 70% of the total attribution, which aligns with common understanding of crash causation. While the average contributions of road, environment, traffic signal, and vehicle factors are generally lower, their distributions reveal considerable variability across individual cases. This suggests that these factors, although not dominant overall, play important roles in specific crashes and should not be overlooked in localized safety analyses. It is also important to note that expert annotations may involve subjectivity, particularly in the precise numerical allocation of scores. However, this challenge can be addressed through appropriate evaluation metrics that emphasize relative consistency rather than exact values, as discussed in Section 3.1.3. Furthermore, in many crashes, multiple contributing factors are present simultaneously. In such cases, the presence of mixed influences tends to dilute the proportion attributed to any single factor, even though each still carries diagnostic value for understanding the conditions that led to the crash.

3.1.2. Root cause attribution modeling

Root cause attribution is formulated as a mechanism-level interpretation task rather than a statistical causal inference problem. The objective is to estimate the relative contributions of predefined contributing factors by explicitly reasoning over contextual evidence contained in crash narratives and diagrams.

Vision-language models are not treated as black-box predictors of causal labels. Instead, they function as rule-guided reasoning agents that apply expert-defined annotation principles to multimodal crash information. As illustrated in Fig. 4, the model is instructed to extract key behavioral, spatial, and contextual cues and organize them into a keyword-guided reasoning chain, in which immediate crash-triggering behaviors (Driver loss of control, Adverse road condition) are distinguished from underlying contextual constraints (Tire or vehicle damage, Ramp curve geometry) and jointly synthesized into factor-level contributions.

These immediate and underlying causal elements are jointly considered to assess how multiple layers of causation collectively contribute to the crash outcome. Based on this integrated reasoning process, each crash is decomposed into five predefined contributing factors, including human, road, environment, vehicle, and traffic signal, with proportional scores reflecting their relative causal strength.

To enforce consistency with expert judgment, the annotation

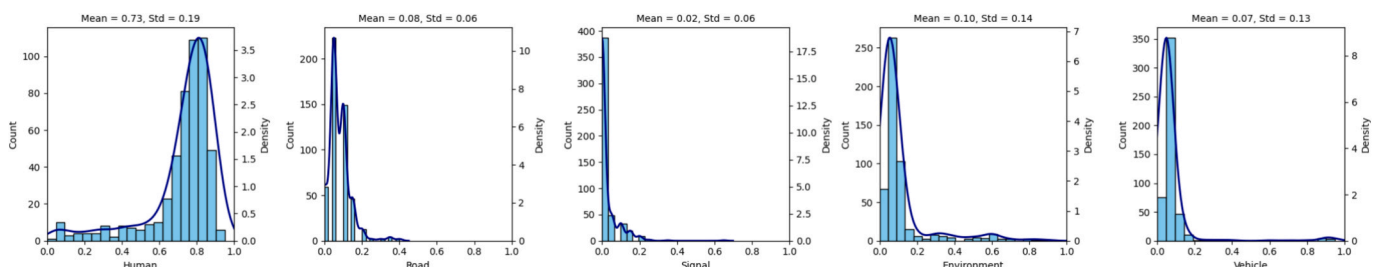


Fig. 3. Distribution of Root Cause Attribution Scores Across 500 Annotated Crash Reports.

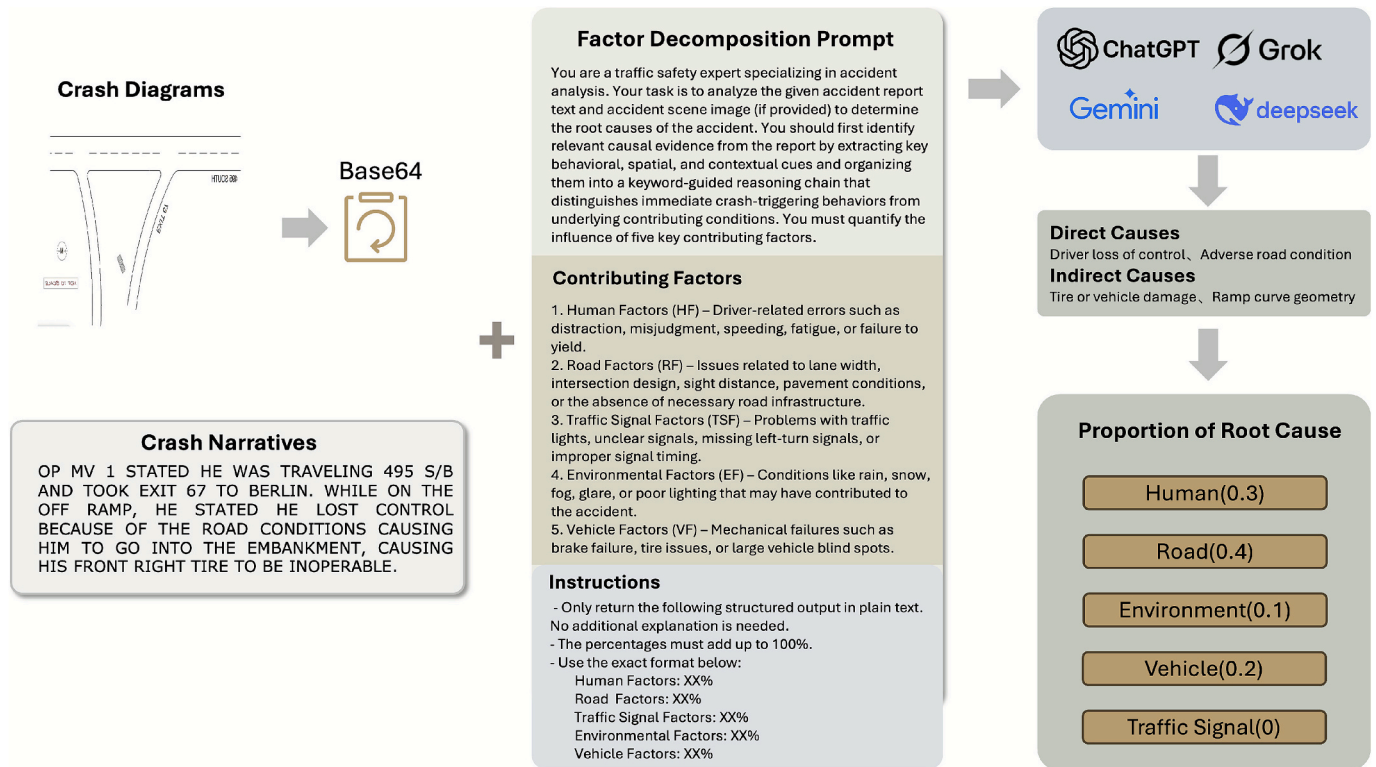


Fig. 4. Structured Prompting Pipeline for Root Cause Decomposition Using Vision-language Model.

principles and proportional scoring rules are explicitly encoded into structured prompts, which require the model to generate intermediate reasoning steps prior to producing a final normalized attribution vector. Under this design, the causal or diagnostic meaning of the attribution results arises from the expert-defined framework, the constrained multimodal input, and the structured reasoning protocol, rather than from unconstrained model behavior. This approach enables scalable and interpretable mechanism-level attribution across large crash datasets. The structured prompting pipeline is illustrated in Fig. 4.

To examine how the proposed mechanism-level attribution framework performs across different large-scale foundation models, we evaluate it using four representative large language or vision-language models: GPT-4o (OpenAI), Grok 2 (xAI), Gemini 2.0 Flash (Google) and DeepSeek V3 (DeepSeek). These models were selected because they are widely adopted in both research and applied settings, provide stable public APIs, and represent diverse architectural designs, training paradigms, and multimodal reasoning capabilities. Evaluating across this set allows us to assess the robustness and generalizability of the proposed framework under different model assumptions and reasoning behaviors.

In addition to foundation-model-based evaluation, several locally implemented models are constructed as benchmark baselines. In these baseline experiments, the attribution task is formulated as a supervised multi-output regression problem, where the target output is a five-dimensional vector representing the proportional contribution of each predefined crash factor. Both text-only and text-and-image inputs are considered to assess the added value of diagram information.

The annotated dataset is divided into training, validation, and testing sets using a 7:1:2 ratio, and this split is applied consistently across all baseline experiments. Three representative model architectures are implemented, including a multimodal ResNet18 and BERT-based fusion model (Ren, 2024), a BLIP-2 based multimodal model (Li et al., 2023), and a text-only BERT-based regression model (Devlin et al, n.d.). These baseline models are used exclusively for comparative evaluation and do not constitute part of the proposed attribution framework.

3.1.3. Evaluation metrics

Since both the model-generated outputs and expert annotations represent proportional attributions of crash causes, evaluating their consistency is essential. While expert annotations are informed by domain knowledge, they are also influenced by individual subjectivity, particularly in the exact numerical assignment of attribution scores. Therefore, model performance should not be judged solely based on precise numerical agreement. Instead, emphasis should be placed on the relative structure of the attribution distribution. Accordingly, the consistency analysis focuses on the agreement between model-predicted attribution vectors and expert-annotated reference vectors rather than pointwise numerical matching.

In expert assessments, the absolute score assigned to a specific factor may vary from one individual to another. For example, one expert might assign a human factor score of 90% to a given crash, while another might assign 80% to the same case. These differences reflect subjective interpretation but do not necessarily indicate disagreement. What offers more meaningful insight is the relative ranking of contributing factors within a single crash and the comparative strength of the same factor across multiple crashes. For instance, in crashes primarily influenced by human behavior, most experts consistently rate the human factor higher than any other. Similarly, when comparing a crash driven by human factors to one influenced by environmental conditions, the human attribution in the former should be higher than in the latter. This perspective is consistent with established statistical reasoning based on within-group and between-group comparisons. Formally, each crash is represented by a five-dimensional continuous attribution vector corresponding to human, vehicle, road, environment, and traffic signal factors, and similarity is assessed based on how well these vectors align between model outputs and expert annotations.

Building on this perspective, we evaluate the model's effectiveness by assessing how well it preserves relative relationships both within individual crashes and across multiple crashes. Accordingly, we adopt two categories of evaluation metrics: crash-level metrics and factor-level metrics. Crash-level metrics evaluate the agreement of attribution

structures within individual crashes, while factor-level metrics assess the consistency of each contributing factor across the entire set of crashes (Table 1).

Crash-level metrics assess the internal consistency of factor ranking and attribution structure within each crash, focusing on whether the model correctly identifies dominant contributors and preserves overall attribution patterns. Factor-level metrics evaluate how well the model captures variation trends of individual contributing factors across the entire crash dataset. For completeness and reproducibility, full metric definitions, mathematical formulations, and additional auxiliary analyses are reported in the Appendix.

3.2. Stage 2: diagnosis of high-risk road locations using street-view imagery

3.2.1. DBSCAN-based crash location clustering

To spatially aggregate crash records for subsequent image-based diagnosis, we applied the Density-Based Spatial Clustering of Applications with Noise (DBSCAN) algorithm (Ester et al., 1996) to all 4,302 geocoded crash locations. DBSCAN is a density-based clustering method that groups together spatially adjacent points without requiring the number of clusters to be specified in advance. It identifies clusters based on local point density and is well suited for identifying irregularly shaped spatial patterns.

Formally, given a point set $D \subset \mathbb{R}^2$, a point $p \in D$ is defined as a core point if the number of points within its ϵ -neighborhood satisfies:

$$|\{q \in D | \text{dist}(p, q) \leq \epsilon\}| \geq \text{minPts} \quad (1)$$

Two points p and q are density-connected if there exists a sequence of core points $p_1, p_2, \dots, p_n$ such that $p_1 = p$, $p_n = q$, and each pair satisfies:

$$\text{dist}(p_i, p_{i+1}) \leq \epsilon \quad (2)$$

A cluster is then defined as the maximal set of points that are density-connected to each other. In our implementation, each crash location is represented by its latitude and longitude. We set the radius parameter ϵ to 50 m and the minimum number of points per cluster minPts to 2. These settings allow the clustering of crashes that occur within a 50-meter range, which is a suitable scale for capturing intersection-level spatial patterns and for ensuring that the corresponding Google Street View images provide sufficient contextual information.

To ensure that all crash locations are retained for analysis, we apply a post-processing adjustment. Specifically, crash points that do not belong to any cluster under the DBSCAN criteria are not discarded as noise. Instead, they are preserved as individual locations. This strategy

prevents the exclusion of potentially meaningful isolated crashes and ensures comprehensive spatial coverage in downstream analysis.

As a result, each DBSCAN cluster, together with each retained single-point location, serves as a spatial anchor for retrieving street-level imagery and conducting road risk assessment.

3.2.2. Road-related risk estimation for clustered locations

To evaluate the road-related risk of each spatial cluster, we define a scoring function for cluster i , which jointly considers the average road attribution within the cluster and the number of crashes it contains. The risk score is formulated as:

$$F(\alpha, \gamma) = \left(\frac{1}{N_i} \sum_{j=1}^{N_i} p_{ij} \right)^\gamma \bullet N_i^\alpha \quad (3)$$

Where N_i is the number of crashes in cluster i , $p_{ij} \in [0, 1]$ is the road attribution score for the j -th crash in cluster i , α is a parameter that controls the influence of crash count, γ is a parameter that adjusts the emphasis on road attribution intensity.

The formula consists of two interpretable components. The first term represents the average level of road-related attribution within the cluster. The exponent γ allows the model to amplify or down-weight the influence of this factor. A higher γ increases the importance of clusters where road infrastructure is frequently identified as a contributing cause. The second term reflects the volume of crash occurrences in the cluster. When α is set to a higher value, clusters with more crashes contribute more strongly to the overall risk ranking.

In this study, we set $\alpha = 0$ and $\gamma = 1$, so that the risk score for each cluster is based solely on the average road attribution proportion, without considering crash frequency. This setting aligns with our goal of identifying locations with pronounced road-related issues, rather than those with simply higher crash counts. By excluding frequency, we ensure that even clusters with fewer crashes can be prioritized if the road factor is consistently high. This allows for a fair comparison focused on road-related risk intensity across all locations.

3.2.3. Street-view-based road condition diagnosis

Based on the road-related risk scores computed in 3.2.2, we select the top $k = 20$ high-risk clusters for further analysis. For each selected cluster, we extract its centroid coordinates and retrieve the corresponding street-view imagery using the Google Maps Street View API. To ensure temporal consistency between crash reports and visual data, street-view images are selected by prioritizing those captured in the same year as the crash occurrence, or as close as possible when exact matches are unavailable. These images provide direct visual evidence of road conditions at each location.

Because multiple street-view images are often available for nearby locations, an image quality screening procedure is applied to exclude images affected by severe backlighting, poor exposure, nighttime conditions, or occlusions. When such issues are detected, the nearest alternative image with higher visual quality is selected, leveraging the spatial redundancy of street-view data.

To perform structured road condition assessment, we use two Vision-language models: Grok 2 and GPT-4o. Each model receives a multimodal input that includes (1) the street-view image of the cluster location, (2) a structured prompt designed to guide infrastructure diagnosis, and (3) a concatenated summary of crash narratives associated with all crashes in the cluster. This combined input allows the model to analyze both the physical roadway conditions and the recurring crash patterns at the location.

The prompt instructs the model to act as a professional traffic safety engineer and evaluate the scene across the following five dimensions:

- **Roadway alignment:** curve sharpness, slope, and sight distance
- **Pavement markings:** clarity, correctness, and completeness

Table 1

All evaluation metrics used in this study, their analytical levels, and their respective roles.

Category	Metric	Evaluation Level	Interpretation
Rank-based	Spearman correlation	Crash-level	Measures consistency of factor ranking within individual crashes
Rank-based	Weighted Spearman	Crash-level	Emphasizes agreement on top-ranked (dominant) factors
Rank-based	Top-k rank match rate	Crash-level	Evaluates whether dominant contributing factors are correctly identified
Distribution-based	Jensen-Shannon distance	Crash-level	Quantifies similarity between predicted and annotated attribution distributions
Correlation-based	Pearson	Factor-level	Measures trend consistency of factor attribution scores across crashes
Distribution-based	Wasserstein distance	Factor-level	Captures global distributional differences for individual factors

- **Road signs:** visibility, legibility, and appropriateness of regulatory and warning signs, such as speed limits and lane merge indicators
- **Surface conditions:** presence of potholes, cracks, and uneven pavement
- **Shoulders and guardrails:** presence, width, continuity, and structural integrity

Each model generates a structured output with two components. A list of observed roadway defects, categorized by infrastructure element and actionable improvement recommendations based on standard engineering practices. To prevent misinterpretation, the input includes safety guidance that directs the model to focus only on physical roadway features and disregard overlaid digital text such as street names commonly present in street view imagery. This ensures that markings identified by the model reflect actual road conditions rather than visual artifacts.

This process, illustrated in Fig. 5, enables a transition from statistical risk identification to interpretable, image-based diagnostics, ultimately supporting targeted interventions at high-risk locations.

To evaluate the quality of road diagnostic outputs generated by the Vision-language models for the top 20 high-risk locations, we conducted a consistency analysis against expert-annotated reference diagnoses. The goal was to assess whether the model-generated observations and recommendations align with human expert interpretations of the same street-view imagery and crash narratives.

To evaluate the quality of the vision-language model generated diagnostic texts, we compared them with expert-written references across five dimensions: roadway alignment, pavement markings, road signs, surface conditions, and shoulders and guardrails. Traditional metrics such as BLEU (Papineni et al., 2002) and ROUGE (Lin, 2004), which rely on exact word or phrase overlap, are not well-suited for this task. These metrics focus on surface-level similarity and penalize variations in wording, even when the generated content is semantically accurate. In contrast, BERTScore provides a better fit for diagnostic generation tasks by measuring semantic similarity between the candidate and reference texts (Zhang et al., 2019).

BERTScore compares each token in the model-generated sentence $C = \{c_1, \dots, c_n\}$ to each token in the expert-written $R = \{r_1, \dots, r_m\}$, computing:

$$Precision = \frac{1}{n} \sum_{i=1}^n \max_j \text{sim}(c_i, r_j) \quad (4)$$

$$Recall = \frac{1}{m} \sum_{j=1}^m \max_i \text{sim}(r_j, c_i) \quad (5)$$

$$F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (6)$$

where $\text{sim}(c_i, r_j)$ denotes the cosine similarity between the contextual embeddings of token c_i in the candidate and token r_j in the reference, both obtained from a pre-trained BERT model. Cosine similarity ranges from -1 to 1 and quantifies how close two-word embeddings are in meaning.

4. Results and discussion

4.1. Root cause attribution result

4.1.1. Comparison of large language models and local models on the test set

To evaluate model performance on crash root cause attribution, we conducted experiments on the held-out test set consisting of 100 annotated crash reports. Table 2 presents comparison results across five crash-level metrics: Spearman correlation, Top-1 accuracy, Top-3 accuracy, Jensen–Shannon distance (JSD), and Weighted Spearman correlation (WSP), for both locally trained baseline models (ResNet18 + BERT-based, BLIP-2, BERT-based) and large language models (GPT-4o, Grok 2, Gemini 2.0 Flash, and DeepSeek V3). These metrics measure the consistency between model-predicted attribution scores and expert annotations, focusing on the agreement in the relative ranking and distribution of contributing factors within each individual crash.

Overall, all models exhibited excellent performance in terms of Top-1 accuracy, achieving values above 92%. This indicates that each model reliably identifies the dominant causal factor in most crashes. Notably, human annotators also exhibited a high level of agreement on the dominant factor, with an agreement rate of 0.972. However, considering Spearman correlation, JSD, and Weighted Spearman correlation metrics, Grok 2 Vision, the multimodal model using both narrative and diagram inputs, significantly outperformed all other models. Particularly,

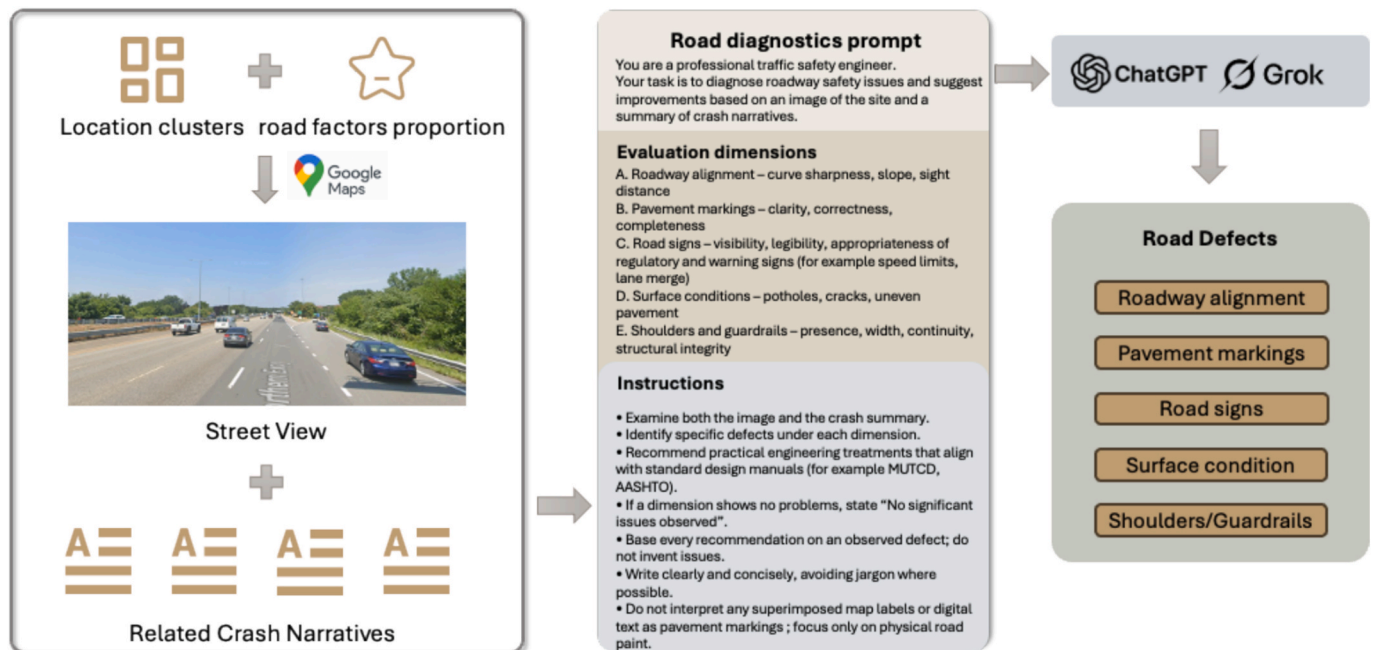


Fig. 5. Structured Road Diagnostics Using Vision-language Model and Street-View Imagery (© Google, imagery date: 2022, modified by authors).

Table 2

Crash-level Metrics Performance on Root Cause Attribution (Test Set, n = 100).

Model	Input	Spearman	Top-1	Top-3	JSD	WSP
ResNet18 + BERT-based	Diagram, Narrative	0.736	0.920	0.747	0.185	0.924
BLIP-2	Diagram, Narrative	0.746	0.920	0.750	0.184	0.923
BERT-based	Narrative	0.737	0.960	0.740	0.170	0.943
GPT-4o	Diagram, Narrative	0.797	0.980	0.813*	0.182	0.952
GPT-4o	Narrative	0.799	0.980	0.790	0.178	0.953
Grok 2	Diagram, Narrative	0.863*	1.000*	0.803	0.131*	0.968*
Grok 2	Narrative	0.828*	1.000*	0.793	0.158*	0.961*
Gemini 2.0 Flash	Narrative	0.736	0.970	0.790	0.193	0.943
DeepSeek V3	Narrative	0.766	0.970	0.750	0.175	0.951

its performance of JSD metric was superior, with pairwise comparisons against all other models yielding a p-value below 0.001, demonstrating that Grok 2 Vision generated attribution distributions most consistent with expert annotations, both in numerical agreement and distribution shape.

In addition to crash-level metrics, we also evaluate model performance using factor-level metrics on the same 100-report test set. These metrics assess how consistently each model predicts the attribution score for a specific factor such as human, vehicle, road, environment and traffic signal, across different crashes. Table 3 presents result for all local models and large language models based on two evaluation criteria: Pearson correlation (γ) and Wasserstein distance (W_1).

All large language models consistently outperform local models on factor-level attribution metrics.

Beyond numerical performance, the superiority of large language models over locally trained regression models can be attributed to their fundamentally different attribution mechanisms. Local models are optimized through supervised learning on imbalanced training data, where human-related causes dominate the majority of samples. As a result, these models tend to systematically overestimate human-factor contributions, even in crashes where roadway, environmental, or vehicle-related factors are clearly dominant.

In contrast, large language models perform attribution through structured semantic reasoning over crash narratives and diagrams rather than frequency-driven parameter fitting. By explicitly extracting and interpreting behavioral, spatial, and contextual cues, these models are less constrained by class imbalance and are better able to preserve non-human attributions in low-frequency but semantically salient cases. This difference is illustrated in Fig. 6, where local models incorrectly inflate human-factor scores, while large language models, particularly Grok 2, correctly identify the dominant non-human causal factors.

Second, large language models leverage extensive general-domain knowledge from their pretraining. Consequently, these models can perform specialized domain analyses effectively through carefully structured prompts without additional training, effectively achieving zero-shot learning. This reduces the necessity for extensive dataset construction and training required by locally trained models.

To examine whether the chosen data split affects model performance

(Hussain et al., 2024), we conducted additional experiments using alternative train/validation/test ratios (60/30/10, 70/10/20, 70/20/10, 80/10/10, and 90/5/5) for locally trained baseline models. Fig. 7 shows that the relative performance of local models remains stable across different splits, with only minor variations in Spearman correlation. In particular, the commonly used 70/10/20 and 70/20/10 settings yield nearly identical results, while splits with smaller test sets exhibit increased variability. These findings indicate that the adopted 7:1:2 split provides a good balance between training sufficiency and test-set stability. Since large language and vision-language models operate in a zero-shot setting without parameter training, their performance is unaffected by data partitioning and therefore not included in this sensitivity analysis.

4.1.2. Validation and in-depth comparison of large language models on the 500-sample set

The previous analysis demonstrated that large language models significantly outperformed locally trained regression models on the 100-report test set. To further validate the relative performance of different large language models, we extend the evaluation to the full set of 500 crash reports with expert-annotated attribution scores. One key advantage of using large language models is that, as zero-shot learners, they do not require fine-tuning or training on a subset of the labeled data. This allows us to utilize the entire annotated dataset for evaluation, enabling more robust and generalizable comparisons. Table 4 presents crash-level evaluation results for all large language models using the same five metrics.

The Grok 2 multimodal model significantly outperforms all other large language models on Spearman correlation, Jensen-Shannon distance (JSD), and Weighted Spearman correlation (WSP). It also achieves consistently high Top-1 and Top-3 accuracy, indicating strong agreement with expert annotations in both factor ranking and distribution. These results validate the earlier findings from the 100-sample test set and confirm that Grok 2 remains the best-performing model on the full 500-report dataset.

A closer examination of attribution behavior reveals that Grok 2 differs from other large language models not only in overall accuracy but also in how causal evidence is synthesized. In crashes involving multiple

Table 3

Factor-Level Attribution Performance of Local Regression Models and Large Language Models (Test Set, n = 100).

Model	Input	Human		Road		Traffic Signal		Environment		Vehicle	
		γ	W_1	γ	W_1	γ	W_1	γ	W_1	γ	W_1
ResNet18 + BERT-based	Diagram, Narrative	0.209	0.084	0.019	0.026	0.446	0.016	0.209	0.059	0.064	0.029
BLIP-2	Diagram, Narrative	0.177	0.081	0.109	0.021	0.453	0.016	0.133	0.059	0.039	0.030
BERT-based	Narrative	0.661	0.041	0.059	0.020	0.292	0.025	0.848	0.031	0.360	0.029
GPT-4o	Diagram, Narrative	0.844*	0.073	0.384*	0.027	0.689*	0.004	0.856	0.030	0.804*	0.072
GPT-4o	Narrative	0.840*	0.077	0.474*	0.029	0.743*	0.005	0.805	0.032	0.859*	0.071
Grok 2	Diagram, Narrative	0.941*	0.028	0.586*	0.026	0.752*	0.008	0.965*	0.015	0.926*	0.023
Grok 2	Narrative	0.907*	0.033	0.467*	0.026	0.704*	0.008	0.932*	0.017	0.894*	0.020
Gemini 2.0 Flash	Narrative	0.771	0.044	0.452*	0.026	0.729*	0.004	0.762	0.061	0.797*	0.066
DeepSeek V3	Narrative	0.832*	0.053	0.372	0.016	0.451	0.008	0.834	0.052	0.901*	0.009

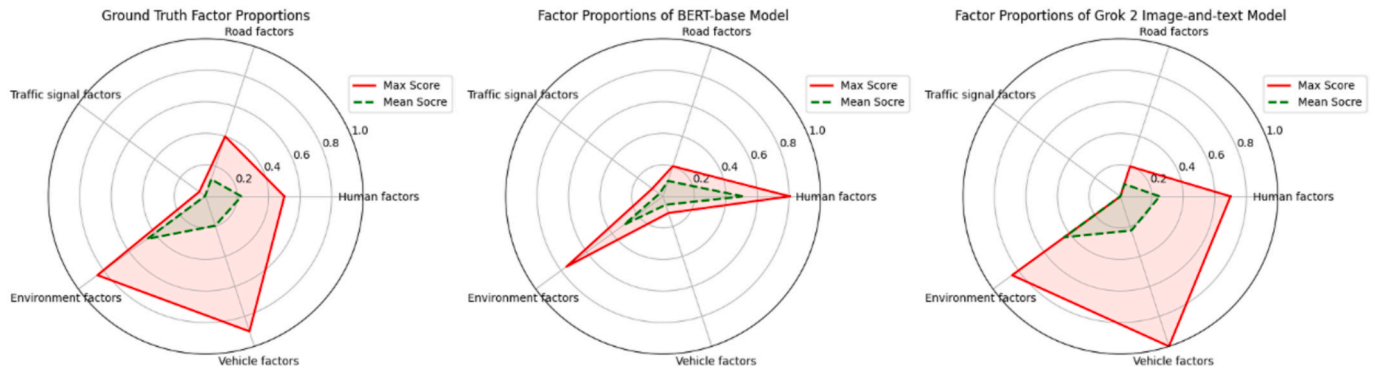


Fig. 6. Attribution Distributions for Non-Human-Dominated Crashes: Ground Truth vs. Model Predictions.

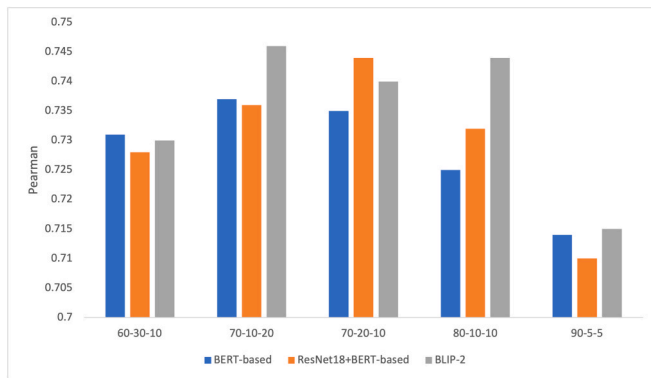


Fig. 7. Sensitivity of Local Model Performance to Data Split Ratios.

Table 4
Crash-Level Attribution Performance of Large Language Models (Annotated Set, $n = 500$).

Model	Input	Spearman	Top-1	Top-3	JSD	WSP
GPT-4o	Diagram, Narrative	0.787	0.958	0.804	0.177	0.949
GPT-4o	Narrative	0.791	0.958	0.801	0.184	0.948
Grok 2	Diagram, Narrative	0.864*	0.982	0.817	0.127*	0.964*
Grok 2	Narrative	0.843	0.984	0.812	0.144	0.961
Gemini 2.0 Flash	Narrative	0.738	0.960	0.815	0.193	0.941
DeepSeek V3	Narrative	0.779	0.970	0.779	0.165	0.953

contributing factors, Grok 2 more consistently distinguishes immediate crash-triggering behaviors from underlying contextual constraints and integrates them into proportional factor-level attribution.

While other models tend to collapse attribution toward a single dominant explanation when faced with ambiguous or mixed evidence, Grok 2 preserves secondary factor contributions that are supported by diagram geometry or contextual descriptions. This behavior leads to more stable attribution distributions and directly contributes to its superior performance in distribution-sensitive metrics such as JSD and WSP.

A comparison between input modalities further reveals that the Grok 2 model shows significantly better performance in its multimodal form than in its text-only variant, particularly in Spearman, JSD, and WSP metrics. In contrast, GPT-4o exhibits similar performance across both input settings. These results suggest that the impact of visual input on attribution outcomes varies across models, with Grok 2 demonstrating a higher degree of sensitivity to image information than GPT-4o.

To further investigate the limitations of the Grok 2 model, we analyzed the bottom 10 cases ranked by Spearman correlation, as shown in Fig. 8. These low-performing examples shared a common characteristic: all were dominated by human-related factors, with attribution proportions exceeding 80%, while the remaining factors held minimal shares. In such scenarios, even minor changes in the ranking of low-weighted factors can lead to substantial drops in Spearman scores, despite the overall attribution being reasonable.

However, in practical applications, accurately identifying dominant factors is typically more valuable than precisely ordering low-impact ones. To better reflect this priority, we examined the Weighted Spearman (WSP) scores for the same set of cases. The WSP metric, which emphasizes high-attribution components, remained consistently high (all above 0.85), suggesting that the model still aligned well with expert judgment where it matters most. This highlights the utility of weighted evaluation metrics in capturing real-world performance more faithfully.

We further evaluate factor-level attribution performance on the full 500-report annotated dataset. The results are summarized in Table 5. Across all five contributing factors, the Pearson correlations between model-generated scores and expert annotations are highly significant for all large language models, with p-values below 0.0001. This indicates that the predicted attribution values are strongly aligned with the ground truth across the entire dataset.

In terms of model-to-model comparisons, the Grok 2 multimodal model consistently outperforms other large language models, achieving the highest average Pearson correlation and lowest Wasserstein distance. These differences are statistically significant across most factors. As shown in Table 5, Grok 2 outperforms all other models in attributing human, environment, and vehicle-related causes, where its predictions align more closely with expert annotations. For road and traffic signal factors, although Grok 2 still yields the highest scores, the performance differences compared to other large language models are not statistically significant. This suggests that when the task involves estimating the individual contribution of a single factor, Grok 2 is particularly reliable for modeling crashes related to human, environmental, or vehicle-related causes. In contrast, for road and traffic signal attribution, all tested large language models provide comparably strong and consistent results, making them equally viable choices for these factors. Based on these results, Grok 2 multimodal model is employed to process the remaining unlabeled data in the full dataset.

4.1.3. Relationship between root cause keywords and human-road attribution scores

To further examine whether the attribution results reflect meaningful causal structure rather than direct score prediction, we analyze the relationship between extracted root cause keywords and the corresponding human- and road-factor attribution scores.

Fig. 9 visualizes the centroids of root cause keywords projected onto a two-dimensional Human-Road attribution space. Each point represents a keyword cluster derived from the model-generated reasoning

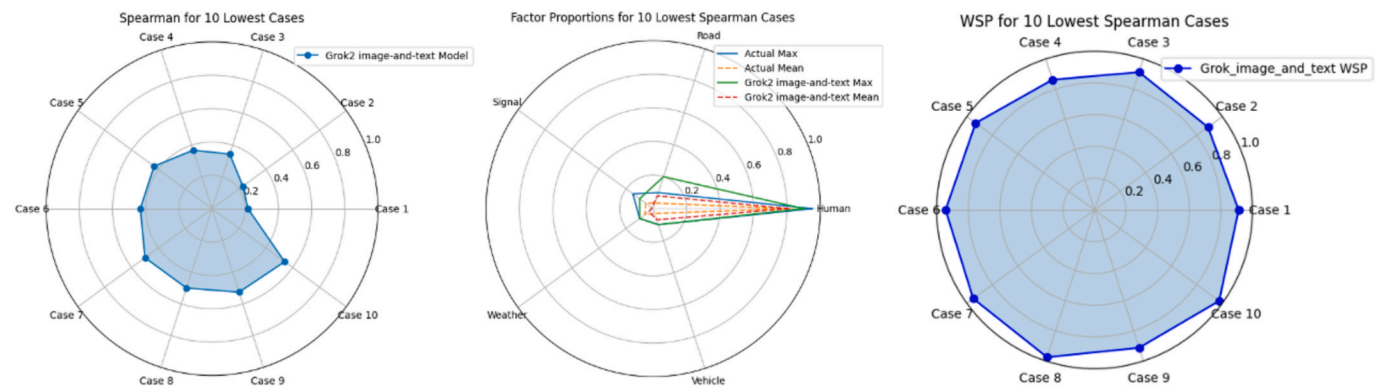


Fig. 8. Error Analysis of the Ten Lowest Spearman Cases for the Grok 2 Multimodal Model.

Table 5
Factor-Level Attribution Performance of Large Language Models (Annotated Set, n = 500).

Model	Input	Human		Road		Traffic Signal		Environment		Vehicle		Mean	
		γ	W_1	γ	W_1	γ	W_1	γ	W_1	γ	W_1	γ	W_1
GPT-4o	Image, Text	0.852	0.075	0.495	0.026	0.771	0.005	0.855	0.019	0.827	0.066	0.760	0.038
GPT-4o	Text	0.822	0.094	0.521	0.028	0.831	0.005	0.833	0.018	0.781	0.084	0.758	0.046
Grok 2	Image, Text	0.911	0.029	0.662	0.023	0.858	0.009	0.931	0.012	0.928	0.023	0.858	0.019
Grok 2	Text	0.885	0.029	0.561	0.022	0.829	0.010	0.902	0.014	0.896	0.020	0.815	0.019
Gemini 2.0 Flash	Text	0.813	0.036	0.502	0.028	0.838	0.010	0.775	0.050	0.760	0.067	0.738	0.038
DeepSeek V3	Text	0.839	0.038	0.571	0.012	0.705	0.012	0.855	0.035	0.884	0.010	0.771	0.021

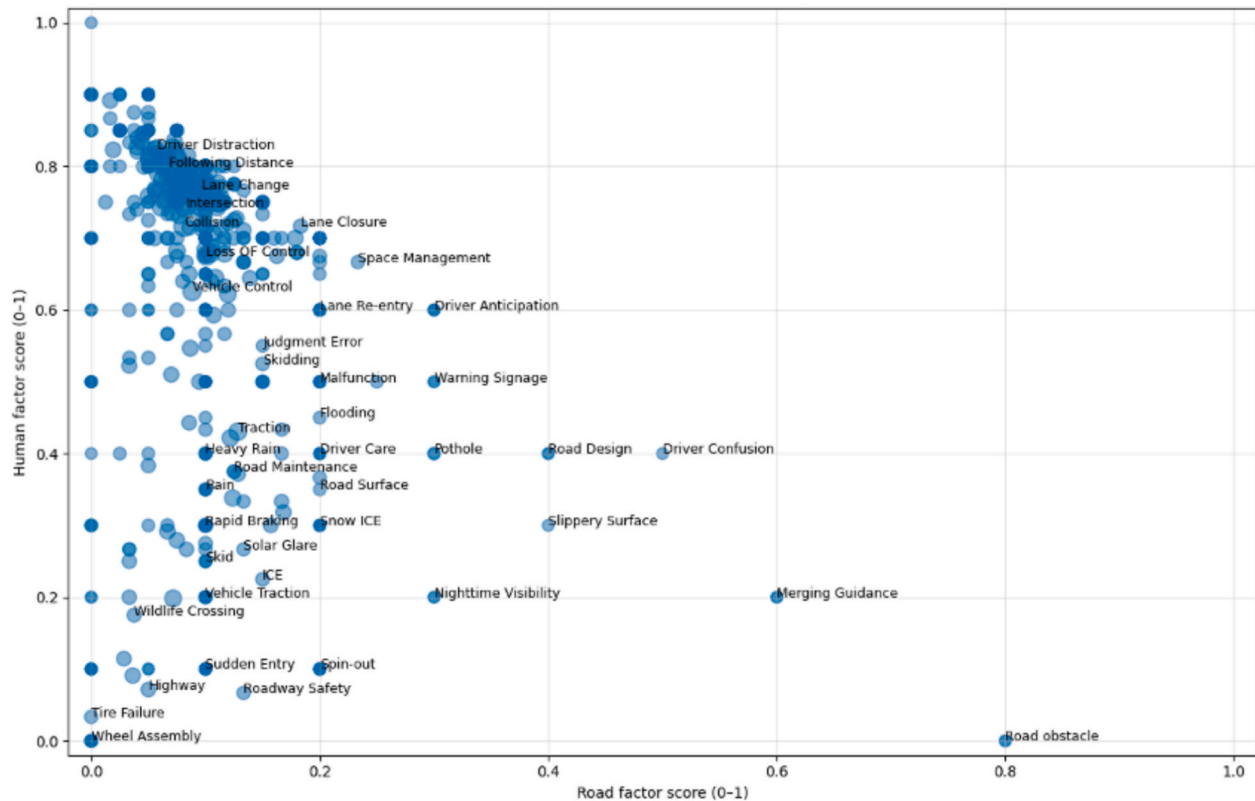


Fig. 9. Distribution of Root Cause Keywords in the Human–Road Attribution Space.

chains, positioned according to the average human-factor and road-factor scores of crashes in which the keyword appears.

Clear semantic patterns emerge from this distribution. Keywords describing driver behaviors, such as “driver distraction”, “lane change” and “loss of control” are concentrated in regions with high human-factor

scores and relatively low road-factor scores. In contrast, keywords associated with roadway characteristics, including “road design,” “merging guidance,” “slippery surface,” and “road obstacle,” exhibit higher road-factor scores and lower human attribution.

Importantly, several keywords occupy intermediate regions of the

space, reflecting joint causal influence. Terms such as “skidding”, “driver confusion” and “vehicle control” are associated with both elevated human and road scores, indicating that the model captures their dual role as immediate behavioral manifestations constrained by roadway or surface conditions.

This analysis demonstrates that factor attribution scores are not directly predicted as abstract labels but instead emerge from the aggregation of keyword-level causal evidence. The alignment between keyword semantics and factor score distributions provides additional validation that the proposed framework performs mechanism-consistent decomposition rather than surface-level classification.

4.2. Diagnosis of locations with elevated road-related risk

4.2.1. Identifying spatial hotspots with high road attribution

To locate spatial hotspots with elevated road-related risk, we first extracted and mapped the geographic coordinates of all 4,302 crash reports. Each report was geocoded using its latitude and longitude. We then applied the DBSCAN algorithm to group crash locations into spatial clusters. A radius of 50 m and a minimum of two points were used to define cluster membership. The 50 m radius corresponds to the typical spatial extent of ramp terminals, merging areas, and channelized conflict zones, and also aligns with the effective visual coverage range of street-view imagery used in downstream diagnostics. Crash points that did not meet the clustering criteria were retained as individual analysis units rather than being discarded as noise, ensuring full spatial coverage for downstream risk assessment and diagnostics.

As a result, the 4,302 crash locations were aggregated into 440 spatial analysis units, including both multi-crash clusters and retained individual points. The spatial clustering outcome is illustrated in Fig. 10 (left).

Based on the root cause attribution results generated by the Grok 2 multimodal model for all 4,302 crash reports, we computed a road-related risk score for each spatial location using the scoring formula introduced in Equation 3. The calculation was performed with parameters $\alpha = 0$ and $\gamma = 1$, which yields the average road factor attribution across all crashes within each cluster or individual point. This configuration intentionally decouples road-related risk intensity from crash frequency, allowing locations with consistently high roadway contribution to be highlighted even when crash counts are relatively low. The resulting spatial distribution of road-related risk is shown in Fig. 10 (right). The resulting risk scores were used to rank all locations, and the top 20 highest-risk hotspots were identified for further street-view-based diagnostics.

4.2.2. Street-view-based road diagnostics

To support interpretable and actionable diagnostics of high-risk locations, we utilized Grok 2 multimodal model to analyze both street-view imagery and crash narratives for each identified hotspot. The model is prompted to simulate the reasoning of a traffic safety engineer and produce structured assessments across five critical roadway design dimensions: roadway alignment, pavement markings, road signs, surface conditions, and shoulders and guardrails. For each dimension, the model identifies potential deficiencies based on visual cues and textual context, and further provides targeted improvement suggestions that align with standard roadway safety guidelines.

To assess the semantic quality of the model-generated road diagnostics, we computed BERTScore precision, recall, and F1 values for each of the five diagnostic dimensions. BERTScore compares the semantic similarity between model outputs and expert-written diagnoses. As shown in Table 6, the Grok 2 multimodal model achieved slightly higher F1 scores than GPT-4o across all dimensions, with an overall mean F1 of 0.796 compared to 0.790 for GPT-4o. These results suggest that Grok 2's outputs are semantically well-aligned with expert-written diagnoses. The model can capture the key meanings conveyed by human assessors, even when expressed in different wording.

Fig. 11 presents an example of this diagnostic process. The input includes the street-view image of the crash site and the associated narrative from a 2022 crash report. Due to the identified issues of ambiguous lane merging and lack of signage, the model recommends enhanced pavement markings and the addition of merge warning signs to improve driver awareness.

Importantly, the roadway signage illustrated in the bottom panel corresponds to the same crash location but is situated upstream of the crash point, where advance guidance is provided to drivers before entering the merge area. Interestingly, this location was indeed modified after the incident. The bottom-left image shows the upstream signage configuration prior to the crash, while the bottom-right image shows the updated upstream signage implemented after the crash.

Although the before-and-after images are not captured from exactly the same camera viewpoint as the crash scene, they represent the same roadway segment and are functionally connected to the crash mechanism by influencing driver lane choice behavior upstream. Rather than directly implementing the model's suggested pavement changes, the actual intervention involved upstream signage enhancements that more clearly indicate lane destinations, thereby encouraging earlier lane changes. This demonstrates that while the model provides valuable diagnostic insight, real-world implementations must also consider broader contextual factors such as roadway geometry, surrounding infrastructure, and construction feasibility. In this case, both the model-

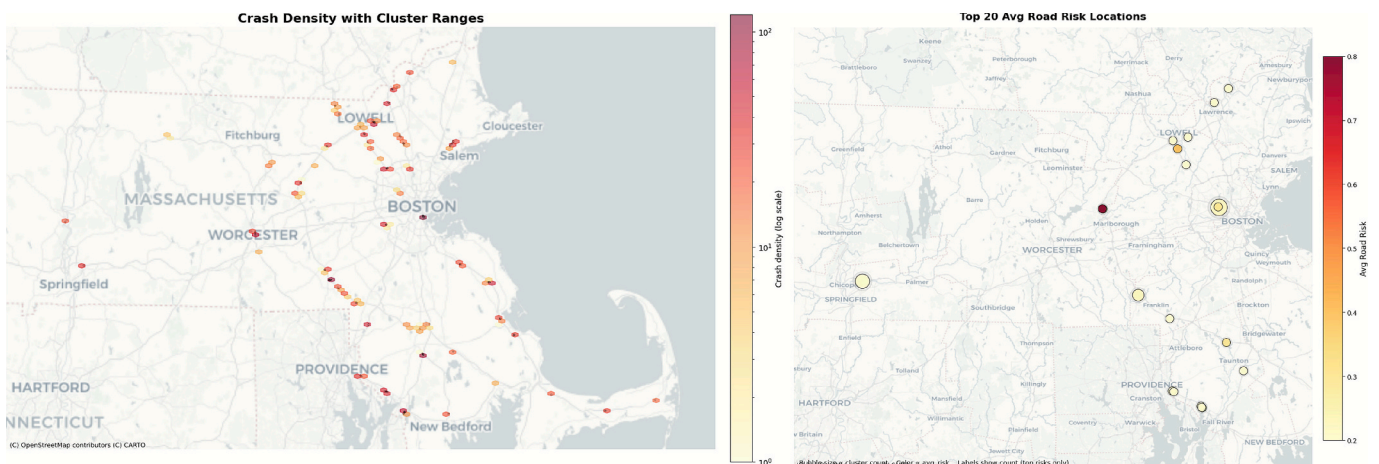


Fig. 10. The spatial clustering results(left) and the distribution of road-related (right).

Table 6
BERTScore Evaluation of Vision-Language Model Generated Road Diagnostics Across Five Dimensions.

Model	Metrix	Roadway Alignment	Pavement Markings	Road Signs	Surface Conditions	Shoulders/ Guardrails	Mean
Grok 2	Precision	0.806	0.805	0.808	0.805	0.804	0.806
	Recall	0.803	0.775	0.784	0.793	0.783	0.788
	F1	0.804	0.790	0.796	0.798	0.793	0.796
Gpt-4o	Precision	0.824	0.812	0.801	0.812	0.799	0.810
	Recall	0.781	0.763	0.764	0.782	0.770	0.772
	F1	0.802	0.787	0.782	0.797	0.784	0.790

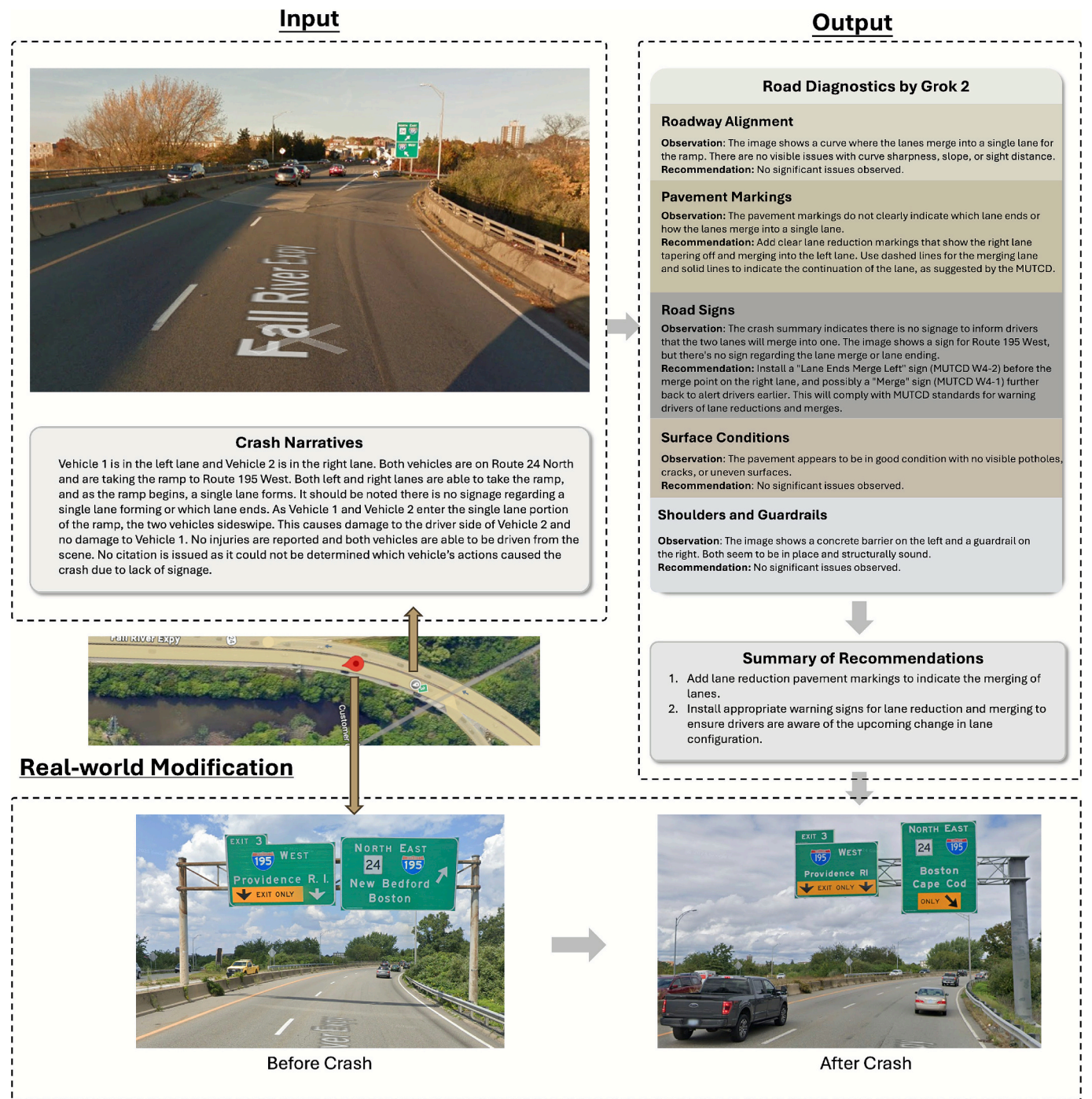


Fig. 11. Example of Multimodal Road Diagnosis Using Grok 2 with Street-View Imagery and Crash Narrative (© Google, imagery dates range from 2021 to 2025, modified by authors).

generated recommendation and the actual redesign aim to reduce driver confusion and mitigate crash risk, albeit through different engineering

approaches.

5. Conclusion and future work

This study introduces a scalable and interpretable framework for automated road safety diagnostics by leveraging Vision-language models to analyze traffic crash reports and street-view imagery. The proposed approach begins with the structured attribution of crash causes into five factor categories based on unstructured textual and visual information in crash reports. Empirical evaluations on a labeled dataset of 500 crash cases demonstrate that large language models substantially outperform traditional regression-based local models in estimating factor contributions, with the Grok 2 multimodal model exhibiting the highest agreement with expert annotations across multiple evaluation metrics.

Building upon this foundation, we identify spatial clusters of high-risk crash locations using DBSCAN and compute road-related risk scores that account for both factor intensity and crash frequency. For the most critical hotspots, the Grok 2 model is used to generate diagnostic insights by interpreting associated street-view images. These generated diagnostics show strong semantic alignment with expert assessments, confirming the model's ability to detect deficiencies in roadway alignment, pavement markings, signage, sight distance, and other safety-related features. Notably, some recommendations align with real-world infrastructure upgrades already implemented, demonstrating the model's potential to support retrospective evaluations and forward-looking planning.

From a policy and engineering practice perspective, the proposed framework provides a mechanism-driven decision support tool that bridges crash analysis and infrastructure intervention. Instead of relying solely on crash counts or aggregate risk indicators, transportation agencies can use factor-level attribution results to distinguish locations where roadway conditions systematically contribute to crash occurrence from those dominated by human behavior. This distinction enables more targeted and defensible investment decisions, such as prioritizing geometric redesign, signage upgrades, or pavement marking improvements at locations where infrastructure plays a recurring causal role.

The proposed framework offers several practical advantages. It supports automated large-scale assessment without requiring site visits, facilitates prioritization of high-risk locations based on both spatial and causal analysis, and provides interpretable diagnostic suggestions that can inform targeted infrastructure improvement. These capabilities make the framework particularly valuable for transportation agencies aiming to optimize the allocation of safety resources. The generated diagnostics can also serve as supporting evidence in safety audits, corridor studies, and project justification processes, thereby strengthening the linkage between data-driven analysis and actionable policy

Appendix

Crash-level metrics assess the internal consistency of factor ranking and distribution within each crash. These include Spearman rank correlation, top-k rank match rate, and Jensen-Shannon distance. To address limitations of standard rank correlation, we additionally propose a weighted Spearman correlation that gives higher importance to the top-ranked factors.

Spearman rank correlation evaluates the consistency of the ranking order between the predicted and ground truth factor attributions. It captures monotonic relationships without assuming linearity and is less sensitive to outliers, making it well-suited for ordinal comparison tasks. The correlation coefficient is computed as:

$$\rho = 1 - \frac{6 \sum_{i=1}^n (r_i - r'_i)^2}{n(n^2 - 1)} \quad (1)$$

where r_i and r'_i represent the ranks of factor i in the ground truth and the predicted attribution vector, respectively, and n is the number of factors. In this study, Spearman correlation is used to assess whether the model correctly reflects the relative importance of factors within individual crashes.

Top-k rank match rate evaluates whether the most important factors identified by the model align with those in the ground truth. It focuses on agreement within the top-k ranked positions, which often correspond to the most critical contributing factors in a crash. The match rate is calculated

implementation.

Despite its promise, the current framework has several limitations. Street view imagery provides a static and often constrained perspective of the roadway environment. Certain crash causes such as temporary obstacles, surface debris, or lighting conditions are time-sensitive and may not be visible in the available images. In addition, fixed camera viewpoints often result in incomplete coverage. At many high-risk locations, only one direction of traffic is visible, making it difficult to assess conditions on the opposite side. Addressing these limitations will require the use of multi-perspective street view imagery or spatially richer data sources such as LiDAR point clouds and aerial drone images to improve diagnostic completeness and accuracy.

Looking ahead, combining street view imagery with vision-language models also holds potential beyond crash causation analysis. For example, it can support large-scale roadway safety audits by automatically identifying key features such as sidewalk continuity, crosswalk visibility, lane configuration, and roadside obstructions across the network. These capabilities may further extend the framework's value in proactive safety assessment and infrastructure planning.

CRediT authorship contribution statement

Guanhe Wu: Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Xintong Yan:** Writing – original draft, Methodology, Data curation, Conceptualization. **Yajie Zou:** Conceptualization. **Yuanchang Xie:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

The data used in this research was obtained from the Massachusetts Department of Transportation. However, the authors are solely responsible for the facts, the accuracy of the data and analysis, and the views presented herein. The authors would also like to thank Dr. Wei Lu of Rockhood.AI for his insightful comments.

Funding

The authors disclosed no financial support for the research, authorship, and/or publication of this article.

as:

$$TopkMatchRate = \frac{|Topk_{true} \cap Topk_{pred}|}{k} \quad (2)$$

where $Topk_{true}$ and $Topk_{pred}$ are the sets of top-k ranked factors in the ground truth and model prediction. This metric emphasizes the model's ability to correctly identify the dominant factors within each crash.

Jensen-Shannon distance (JSD) quantifies the similarity between two probability distributions in a symmetric and smoothed manner. It is a bounded and interpretable divergence metric derived from Kullback–Leibler divergence. The distance is computed as:

$$JSD(P||Q) = \frac{1}{2}KL(P||M) + \frac{1}{2}KL(Q||M) \quad (3)$$

where P and Q are the predicted and true attribution vectors, and $M = \frac{1}{2}(P + Q)$. KL denotes the Kullback–Leibler divergence. JSD is always non-negative and bounded between 0 and 1. A lower JSD value indicates higher similarity between the predicted and annotated attribution distributions. In this study, JSD is used to assess the overall alignment between the model's output and expert-labeled factor proportions within each crash.

Weighted Spearman correlation is an extension of the standard Spearman rank correlation, designed to assign higher importance to agreement at top-ranked positions. In many crash reports, one factor may clearly dominate the attribution (for example, human error), while the remaining factors contribute only marginally. In such cases, the precise ordering among low-impact factors becomes less meaningful for interpretation or policy relevance. The standard Spearman correlation treats all rank positions equally, which may underrepresent the model's ability to correctly identify the most critical factor. To address this, we adopt the weighted version of the metric, which reduces the influence of low-ranked positions and better aligns with the practical emphasis on top-ranked contributors. The weighted correlation is computed as:

$$\rho = 1 - \frac{6 \sum_{i=1}^n w_i (r_i - r'_i)^2}{\sum_{i=1}^n w_i \bullet (n^2 - 1)} \quad (4)$$

where w_i is the weight assigned to rank position i . This metric is particularly useful for evaluating ranking accuracy in imbalanced attribution scenarios, where a small number of dominant factors should be prioritized in evaluation.

Factor-level metrics evaluate how well a specific factor's attribution scores are preserved across different crashes. These metrics assess the consistency of model predictions with expert annotations at a global level.

Pearson correlation measures the linear relationship between the predicted and annotated scores for a given factor across all crashes. It reflects how well the model captures variation trends in factor importance. The coefficient is computed as:

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \bullet \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}} \quad (5)$$

where x_i and y_i are the predicted and true scores for a given factor in crash i , and $\bar{x}, \bar{y}$ are their respective means. A higher correlation indicates better alignment between model outputs and expert scores for that factor.

Wasserstein distance quantifies how different two distributions are by measuring the cost of transforming one into the other. For one-dimensional distributions P and Q , the first-order Wasserstein distance is:

$$W(P, Q) = \int_0^1 |F_P^{-1}(t) - F_Q^{-1}(t)| dt \quad (6)$$

where F_P^{-1} and F_Q^{-1} are the inverse cumulative distribution functions of P and Q . A smaller Wasserstein distance indicates greater similarity between the predicted and true score distributions, capturing both magnitude and distributional shape differences.

Data availability

The authors do not have permission to share data.

References

- Afghari, A.P., Washington, S., Prato, C., Haque, M.M., 2019. Contrasting case-wise deletion with multiple imputation and latent variable approaches to dealing with missing observations in count regression models. *Anal. Methods Accid. Res.* 24. <https://doi.org/10.1016/j.amar.2019.100104>.
- Aguiar, M., Stolzer, A., Boyd, D.D., 2017. Rates and causes of accidents for general aviation aircraft operating in a mountainous and high elevation terrain environment. *Accid. Anal. Prev.* 107, 195–201. <https://doi.org/10.1016/j.aap.2017.03.017>.
- Ali, Y., Hussain, F., Haque, M.M., 2024. Advances, challenges, and future research needs in machine learning-based crash prediction models: a systematic review. *Accid. Anal. Prev.* 194. <https://doi.org/10.1016/j.aap.2023.107378>.
- Al-Mistarehi, B.W., Alomari, A.H., Imam, R., Mashqba, M., 2022. Using machine learning models to forecast severity level of traffic crashes by R studio and ArcGIS. *Front. Built Environ.* 8. <https://doi.org/10.3389/fbuil.2022.860805>.
- Arteaga, C., Oh, I., Park, C., Park, J., 2024. Extraction of traffic crash locations from text narratives: a zero-shot approach using large language models, in: *Proceedings of the 12th IPMA Research Conference "Project Management in the Age of Artificial Intelligence."* International Project Management Association – IPMA, IPMA USA, pp. 180–188. <https://doi.org/10.56889/hwfq5547>.
- Arteaga, C., Park, J., 2025. A large language model framework to uncover underreporting in traffic crashes. *J. Saf. Res.* 92, 1–13. <https://doi.org/10.1016/j.jsr.2024.11.009>.
- Arteaga-Sanchez, C.D., n.d. Identification of factors contributing to traffic crashes by identification of factors contributing to traffic crashes by analysis of text narratives repository citation repository citation. <https://doi.org/10.34917/35777454>.
- Cai, Q., Abdel-Aty, M., Zheng, O., Wu, Y., 2022. Applying machine learning and google street view to explore effects of drivers' visual environment on traffic safety. *Transp. Res. Part C Emerg. Technol.* 135. <https://doi.org/10.1016/j.trc.2021.103541>.
- DeepSeek, 2024. DeepSeek Large Language Model. <https://www.deepseek.com> (accessed January 2026).
- Devlin, J., Chang, M.-W., Lee, K., Google, K.T., Language, A.I., n.d. BERT: pre-training of deep bidirectional transformers for language understanding.
- Ester, M., Krieger, H.-P., Sander, J., Xu, X., 1996. A density-based algorithm for discovering clusters in large spatial databases with noise.

- Fan, Z., Wang, P., Zhao, Yang, Zhao, Yibo, Ivanovic, B., Wang, Z., Pavone, M., Yang, H.F., 2024. Learning traffic crashes as language: datasets, benchmarks, and what-if causal analyses.
- Fitzpatrick, C.D., Rakasi, S., Knodler, M.A., 2017. An investigation of the speeding-related crash designation through crash narrative reviews sampled via logistic regression. *Accid. Anal. Prev.* 98, 57–63. <https://doi.org/10.1016/j.aap.2016.09.017>.
- Gao, H., Zhao, Y., 2025. Application of vision-language model to pedestrians behavior and scene understanding in autonomous driving.
- Gao, J., Sarkar, B., Xia, F., Xiao, T., Wu, J., Ichter, B., Majumdar, A., Sadigh, D., 2024. Physically Grounded Vision-Language Models for Robotic Manipulation. In: *Proceedings - IEEE International Conference on Robotics and Automation*. Institute of Electrical and Electronics Engineers Inc, pp. 12462–12469. <https://doi.org/10.1109/ICRA57147.2024.10610090>.
- Google, 2022. Google Maps Platform: Street View. <https://developers.google.com/maps/documentation/streetview/> (accessed January 2026).
- Google, 2024. Gemini Large Language Model. <https://deepmind.google/technologies/gemini/> (accessed January 2026).
- Grigorev, A., Saleh, K., Ou, Y., Mihaita, A.-S., 2024. Enhancing traffic incident management with large language models: a hybrid machine learning approach for severity classification. <https://doi.org/10.1007/s13177-024-00448-7>.
- Guo, Z., Yagudin, Z., Lykov, A., Konenkov, M., Tsetserukou, D., 2024. VLM-Auto: VLM-based autonomous driving assistant with human-like behavior and understanding for complex road scenes. In: *2024 2nd International Conference on Foundation and Large Language Models, FLLM 2024*. Institute of Electrical and Electronics Engineers Inc, pp. 501–507. <https://doi.org/10.1109/FLLM63129.2024.10852498>.
- Hussain, F., Ali, Y., Li, Y., Haque, M.M., 2024. A bi-level framework for real-time crash risk forecasting using artificial intelligence-based video analytics. *Sci. Rep.* 14. <https://doi.org/10.1038/s41598-024-54391-4>.
- Jain, S., Thapa, S., Chen, K.T., Abbott, A.L., Sarkar, A., 2024. Semantic understanding of traffic scenes with large vision language models, in: *IEEE Intelligent Vehicles Symposium, Proceedings*. Institute of Electrical and Electronics Engineers Inc., pp. 1580–1587. <https://doi.org/10.1109/IV55156.2024.10588373>.
- Kwayu, K.M., Kwigizile, V., Lee, K., Oh, J.S., 2021. Discovering latent themes in traffic fatal crash narratives using text mining analytics and network topology. *Accid. Anal. Prev.* 150. <https://doi.org/10.1016/j.aap.2020.105899>.
- Li, J., Li, D., Savarese, S., Hoi, S., 2023. BLIP-2: bootstrapping language-image pre-training with frozen image encoders and large language models.
- Li, P., Chen, S., Yue, L., Xu, Y., Noyce, D.A., 2024. Analyzing relationships between latent topics in autonomous vehicle crash narratives and crash severity using natural language processing techniques and explainable XGBoost. *Accid. Anal. Prev.* 203. <https://doi.org/10.1016/j.aap.2024.107605>.
- Lin, C.-Y., 2004. ROUGE: a package for automatic evaluation of summaries.
- Liu, Q., Song, P., Li, F., 2025a. Exploring the dynamic determinants of general aviation accidents across flight phases and time: a random parameter bivariate probit approach with heterogeneity in means. *Anal. Methods Accid. Res.* 47. <https://doi.org/10.1016/j.amar.2025.100386>.
- Liu, Y., Chen, T., Chung, H., Jang, K., Xu, P., 2025b. Is there an emotional dimension to road safety? A spatial analysis for traffic crashes considering streetscape perception and built environment. *Anal. Methods Accid. Res.* 46. <https://doi.org/10.1016/j.amar.2025.100374>.
- Mumtarin, M., Chowdhury, M.S., Wood, J., 2023. Large language models in analyzing crash narratives – a comparative study of ChatGPT, BARD and GPT-4. *OpenAI*, 2024. GPT-4o. <https://openai.com> (accessed January 2026).
- Papineni, K., Roukos, S., Ward, T., Zhu, W.-J., 2002. BLEU: a method for automatic evaluation of machine translation.
- Ren, J., 2024. Multimodal sentiment analysis based on BERT and ResNet.
- Ren, M., Zhang, X., Zhi, X., Wei, Y., Feng, Z., 2024. An annotated street view image dataset for automated road damage detection. *Sci. Data* 11. <https://doi.org/10.1038/s41597-024-03263-7>.
- Roque, C., Lourenço Cardoso, J., Connell, T., Schermers, G., Weber, R., 2019. Topic analysis of Road safety inspections using latent dirichlet allocation: a case study of roadside safety in Irish main roads. *Accid. Anal. Prev.* 131, 336–349. <https://doi.org/10.1016/j.aap.2019.07.021>.
- Santos, D., Saías, J., Quaresma, P., Nogueira, V.B., 2021. Machine learning approaches to traffic accident analysis and hotspot prediction. *Computers* 10. <https://doi.org/10.3390/computers10120157>.
- Sayed, M.A., Qin, X., Kate, R.J., Anisuzzaman, D.M., Yu, Z., 2021. Identification and analysis of misclassified work-zone crashes using text mining techniques. *Accid. Anal. Prev.* 159. <https://doi.org/10.1016/j.aap.2021.106211>.
- Shao, Y., Shi, X., Zhang, Y., Shiwakoti, N., Xu, Y., Ye, Z., 2024. Injury severity prediction and exploration of behavior-cause relationships in automotive crashes using natural language processing and extreme gradient boosting. *Eng. Appl. Artif. Intel.* 133. <https://doi.org/10.1016/j.engappai.2024.108542>.
- Tanprasert, T., Siripanpornchana, C., Surasvadi, N., Thajchayapong, S., 2020. Recognizing traffic black spots from street view images using environment-aware image processing and neural network. *IEEE Access* 8, 121469–121478. <https://doi.org/10.1109/ACCESS.2020.3006493>.
- Van, M.H., Verma, P., Wu, X., 2024. On large visual language models for medical imaging analysis: an empirical study, in: *Proceedings - 2024 IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technologies, CHASE 2024*. Institute of Electrical and Electronics Engineers Inc., pp. 172–176. <https://doi.org/10.1109/CHASE60773.2024.00029>.
- Wang, C., Abdel-Aty, M., Han, L., 2025. Grouped random parameters poisson-lindley model with spatial effects addressing crashes at intersections: insights from visual environment features and spatiotemporal instability. *Anal. Methods Accid. Res.* 47. <https://doi.org/10.1016/j.amar.2025.100387>.
- xAI, 2024. Grok-2. <https://x.ai> (accessed January 2026).
- Yan, K., Wang, Y., Jia, L., Wang, W., Liu, S., Geng, Y., 2023. A content-aware corpus-based model for analysis of marine accidents. *Accid. Anal. Prev.* 184. <https://doi.org/10.1016/j.aap.2023.106991>.
- Yan, X., He, J., Zhang, C., Liu, Z., Qiao, B., Zhang, H., 2021a. Single-vehicle crash severity outcome prediction and determinant extraction using tree-based and other non-parametric models. *Accid. Anal. Prev.* 153. <https://doi.org/10.1016/j.aap.2021.106034>.
- Yan, X., He, J., Zhang, C., Liu, Z., Wang, C., Qiao, B., 2021b. Temporal analysis of crash severities involving male and female drivers: a random parameters approach with heterogeneity in means and variances. *Anal. Methods Accid. Res.* 30. <https://doi.org/10.1016/j.amar.2021.100161>.
- Ye, X., Li, S., Gong, W., Li, X., Dadashova, B., Li, W., Du, J., Wu, D., 2025. Street view imagery in traffic crash and road safety analysis: a review. *Appl. Spat. Anal. Policy*. <https://doi.org/10.1007/s12061-025-09653-7>.
- Yu, X., Ma, J., Tang, Y., Yang, T., Jiang, F., 2024. Can we trust our eyes? Interpreting the misperception of road safety from street view images and deep learning. *Accid. Anal. Prev.* 197. <https://doi.org/10.1016/j.aap.2023.107455>.
- Yue, H., 2025. Investigating streetscape environmental characteristics associated with road traffic crashes using street view imagery and computer vision. *Accid. Anal. Prev.* 210. <https://doi.org/10.1016/j.aap.2024.107851>.
- Zhang, T., Kishore, V., Wu, F., Weinberger, K.Q., Artzi, Y., 2019. BERTScore: evaluating text generation with BERT.
- Zhang, X., Zhao, X., Bian, Y., Huang, J., Yin, L., 2024. Interactive effects analysis of road, traffic, and weather characteristics on shared e-bike speeding risk: a data-driven approach. *Accid. Anal. Prev.* 207. <https://doi.org/10.1016/j.aap.2024.107755>.
- Zheng, O., Abdel-Aty, M., Wang, D., Wang, C., Ding, S., 2023. TrafficSafetyGPT: tuning a pre-trained large language model to a domain-specific expert in transportation safety.
- Zhou, X., Liu, M., Yurtsever, E., Zagar, B.L., Zimmer, W., Cao, H., Knoll, A.C., 2024. Vision language models in autonomous driving: a survey and outlook. *IEEE Trans. Intell. Veh.* <https://doi.org/10.1109/TIV.2024.3402136>.